

A REMARK ON GRADIENTS OF HARMONIC FUNCTIONS IN DIMENSION ≥ 3

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INTRODUCTION

The purpose of this note is to refine the result of [2] on the existence of $C^{1+\varepsilon}$ -harmonic functions on $\mathbb{R}_+^d$ ($d \geq 3$) for which ∇f vanishes on a boundary set of positive measure.

Using some new ingredients originating from [1], one actually gets vanishing of $f, \nabla f$ simultaneously (see the Theorem below). The problem whether this may happen for functions of class C^2 or C^∞ remains open (see the remarks at the end). The present construction is based on the same techniques as in [2] but in this case is simpler because the correction theorem is applied to scalar functions rather than vector valued functions. The reader may wish to consult [2] for a motivation and more detailed discussion of this problematic.

Theorem. *If $d \geq 3$ there is a harmonic function $f : \mathbb{R}_+^d \rightarrow \mathbb{R}$ which is C' up to the boundary and such that f and ∇f vanish on a common boundary set with positive measure.*

Outline of proof is as follows. Start with a function u_0 which vanishes on an open subset of the boundary. Using a successive modification ("correction theorem") procedure, decrease the normal derivative to 0 on a subset with large measure. To get the theorem it is necessary to keep the original function unchanged on a set with large measure. Thus the functions added on during the modification procedure must have small compact support. The idea for constructing those functions is due to A.B. Alexandrov - P. Kargaev [1] who used it for a closely related problem.

Lemma 1. *If $p > 0$ is small enough then for all sufficiently small $\varepsilon \exists F_\varepsilon : \mathbb{R}^{d-1} \rightarrow \mathbb{R}$, $\text{supp } F_\varepsilon \subset D(0, \varepsilon^{1/2})$ ($= \{x \in \mathbb{R}^{d-1} : |x| \subset \varepsilon^{1/2}\}$) and (letting $\hat{F}_\varepsilon$ be its harmonic extension to $\mathbb{R}_+^d$)*

$$\int_{\mathbb{R}^{d-1}} \left(\left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - 1 \right) dx < -\eta$$

with $\eta > 0$ independent of ε . Moreover $|\nabla \hat{F}_\varepsilon| \lesssim \min(\varepsilon^{-d}, |x|^{-d})$ on $\mathbb{R}^{d-1}$.

Proof. Alexandrov-Kargaev use the function $G_\varepsilon : \mathbb{R}_+^d \rightarrow \mathbb{R}$ defined by

$$G_\varepsilon(x) = \frac{-(\varepsilon + x_d)}{|x + \varepsilon e_d|^d}$$

(e_d = unit vector in x_d direction) which, as they show, satisfies

$$\int \left(\left| 1 + \frac{dG_\varepsilon}{dn} \right|^p - 1 \right) dx < -2M < 0$$

for p below a certain critical number and ε sufficiently small. One can go from this function to $\hat{F}_\varepsilon$ as above, as follows:

Let $\delta = \varepsilon^{1/2}$ and for $j = 0, 1, 2, \dots$ let $\psi_j = \mathbb{R}^{d-1} \rightarrow \mathbb{R}$ be functions with $\text{supp } \psi_j \subset \{2^{j-1}\delta \leq |x| \leq 2^{j+1}\delta\}$ and $\sum \psi_j = 1$ for $|x| > \delta$, and with the natural bounds i.e. $|\nabla^k \psi_j| \lesssim (2^j \delta)^{-k}$. Let

$\psi = \sum \psi_j$. Let $\rho_\varepsilon^j = \psi_j G_\varepsilon$, $\rho_\varepsilon = \sum_j \rho_\varepsilon^j = \psi G_\varepsilon$. Let $F_\varepsilon = G_\varepsilon - \rho_\varepsilon$. To make the estimates, let ∇_T be differentiation in the $\mathbb{R}^{d-1}$ directions. Then $|\nabla_T^k G_\varepsilon| \leq \varepsilon |x|^{-(d+k)}$, $k = 0, 1, 2$. Therefore $|\nabla_T^k \rho_\varepsilon^j| \lesssim \varepsilon \cdot (2^j \delta)^{-(d+k)}$. This implies $\|\nabla_T \rho_\varepsilon^j\|_1 \lesssim \varepsilon (2^j \delta)^{-2}$, and also $\|\nabla_T \rho_\varepsilon^j\|_{C^\alpha} \lesssim \varepsilon (2^j \delta)^{-(d+1+\alpha)}$. By $L^1 \rightarrow \text{weak } -L^1$ and Hölder estimates for the Riesz transforms we have

$$\left| \left\{ x : \left| \frac{d\hat{\rho}_\varepsilon^j}{dn}(x) \right| > \lambda \right\} \right| < \varepsilon (2^j \delta)^{-2} \lambda^{-1}$$

$$\left\| \frac{d\hat{\rho}_\varepsilon^j}{dn} \right\|_{C^\alpha} \lesssim \varepsilon (2^j \delta)^{-(d+1+\alpha)}$$

It follows that $\left| \frac{d\hat{\rho}_\varepsilon^j}{dn}(x) \right| \lesssim \varepsilon (2^j \delta)^{-(d+1)}$ when $|x| < 2 \cdot 2^{j+1} \delta$ (the weak type 1 estimate implies this holds for *some* x with $|x| < 2 \cdot 2^{j+1} \delta$ and the Hölder estimate extends it to *all* such x). Furthermore if $|x| > 2 \cdot 2^{j+1} \delta$ then

$$\left| \frac{d\hat{\rho}_\varepsilon^j}{dn}(x) \right| = \left| \text{const} \cdot \int_{|y| < 2^{j+1} \delta} |x - y|^{-d} \rho_\varepsilon^j(y) dy \right| \lesssim \varepsilon (2^j \delta)^{-1} |x|^{-d},$$

using $|\rho_\varepsilon^j| \lesssim \varepsilon \cdot (2^j \delta)^{-d}$.

In other words we have the following:

$$\left| \frac{d\hat{\rho}_\varepsilon^j}{dn} \right| \lesssim \varepsilon (2^j \delta)^{-1} \min(|x|^{-d}, (2^j \delta)^{-d}) \quad \text{on } \mathbb{R}^{d-1}.$$

Summing over j we obtain

$$\left| \frac{d\hat{\rho}_\varepsilon}{dn} \right| \lesssim \varepsilon \delta^{-1} \min(\delta^{-d}, |x|^{-d}) \quad (*)$$

Then for small p and appropriate η

$$\begin{aligned} \int \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - 1 &= \left(\int \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p - 1 \right) + \left(\int \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right) \\ &\leq -2\eta + \int_{|x| < 1} \left| \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right| + \int_{|x| > 1} \left(\left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right) \end{aligned}$$

Now

$$\int_{|x| < 1} \left| \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right| \leq \int_{|x| \leq 1} \left| \frac{d\rho_\varepsilon}{dn} \right|^p \quad ((a+b)^p \leq a^p + b^p \text{ when } p < 1)$$

and $(*)$ implies this goes to 0 with ε . When $|x| > 1$ we have a lower hand for $|1 + \frac{dG_\varepsilon}{dn}|$, and $|\frac{d\rho_\varepsilon}{dn}|$ is small. It follows by the mean value theorem that

$$\left| \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right| \leq C \cdot \left| \frac{d\rho_\varepsilon}{dn} \right|$$

and therefore that

$$\int_{|x| > 1} \left| \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right| \leq C \int_{|x| > 1} \left| \frac{d\rho_\varepsilon}{dn} \right| \leq C \varepsilon \delta^{-p}$$

which again goes to 0 with ε . That proves $\int \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - 1 < -\eta$.

To prove $|\nabla \widehat{F}_\varepsilon| \lesssim \min(\varepsilon^{-d}, |x|^{-d})$: this estimate is obvious for $|\nabla G_\varepsilon|$ and for $|\nabla_T \rho_\varepsilon|$. For $|\frac{d\widehat{\rho}_\varepsilon}{dn}|$ it follows from (*). $\square$

Lemma 2. *If N is large enough there is a constant $\beta \subset \beta(N) > 0$ such that: if $Q \subset \mathbb{R}^{d-1}$ is a cube, a_Q its center and $I : Q \rightarrow \mathbb{R}$ is a function such that*

$$N^{d-1} |I(a_Q)|^{-1} \sup_{x \in Q} |I(x) - I(a_Q)|$$

is sufficiently small (independently of N and ε) then

$$\left(\int_Q \left| I(x) + \frac{d\widehat{F}_\varepsilon}{dn} (N\ell(Q)^{-1}(x - a_Q)) \right|^p dx \right)^{1/p} < e^{-2\beta} |I(a_Q)| |Q|^{1/p}.$$

Proof. We can assume $Q = Q(N) = [-\frac{N}{2}, \frac{N}{2}] \times \cdots \times [-\frac{N}{2}, \frac{N}{2}]$. It is then equivalent to show that if $N^{d-1} \|I - 1\|_\infty$ is sufficiently small then

$$\int_{Q(N)} \left| I + \frac{d\widehat{F}_\varepsilon}{dn} \right|^p - 1 \leq -\eta$$

for suitable $\eta > 0$. This last inequality is true since

$$\begin{aligned} \int_{Q(N)} \left| 1 + \frac{d\widehat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{d\widehat{F}_\varepsilon}{dn} \right|^p dx &\leq \int_{Q(N) \setminus D(0,1)} + \int_{D(0,1)} \\ &\leq C \int_{Q(N) \setminus D(0,1)} |I - 1| dx + \int_{D(0,1)} |I - 1|^p dx \end{aligned}$$

(because $1 + \frac{d\widehat{F}_\varepsilon}{dn}$ is bounded away from 0 when $|x| > 1$)

$$\leq C \|I - 1\|_\infty N^{d-1} + C \|I - 1\|_\infty^p,$$

which is small. It remains to apply lemma 1. $\square$

Now we give the recursive construction. For a sufficiently large N to be determined, choose β according to lemma 2. Also fix sequences $K_n \rightarrow \infty$, $\varepsilon_n \searrow 0$ (to be determined). Constants denoted C will be independent of these choices. Let u_0 be a smooth function on $\mathbb{R}^{d-1}$ which vanishes on $Q(1)$.

We will drop the $\widehat{}$ notation and identify functions on $\mathbb{R}^{d-1}$ to their harmonic extensions when there is no confusion.

If the n th stage of the construction has been done ($n \geq 0$) we will have the following data: a number $\delta_n > 0$ with $\delta_n^{-1} \in \mathbb{Z}$, a subset $\mathcal{Q}_n \subset \mathcal{H}_n$ where $\mathcal{H}_n$ is the collection of $\delta_n^{-(d-1)}$ cubes of side δ_n whose union is $Q(1)$, and a smooth function u_n such that

$$\left(\int_{V_n} \left| \frac{du_n}{dn} \right|^p \right)^{1/p} \leq A e^{-\beta n}$$

where $V_n = \bigcup \{Q : Q \in \mathcal{Q}_n\}$, β is as above and A is a sufficiently large constant which is independent of n . To start take $\delta_0 = 1$, $\mathcal{Q}_0 = \{Q(1)\}$.

To do stage $n + 1$ choose δ_{n+1} with $\frac{\delta_n}{\delta_{n+1}} \in \mathbb{Z}$ such that δ_{n+1} is extremely small - how small will be determined later. $\mathcal{Q}_{n+1}$ is then all cubes $Q \in \mathcal{H}_{n+1}$ such that

(a) the cube $Q' \in \mathcal{H}_n$ with $Q \subset Q'$ satisfies $Q' \in \mathcal{Q}_n$, and

(b) $\left(\frac{1}{|Q|} \int_Q \left| \frac{du_n}{dn} \right|^p \right)^{1/p} < K_{n+1} e^{-\beta n}.$

Let a_Q be the center of Q and define

$$u_{n+1}(x) = u_n(x) + \sum_{Q \in \mathcal{Q}_{n+1}} \frac{du_n}{dn}(a_Q) F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q)) \cdot \frac{\delta_{n+1}}{N}$$

so that

$$\nabla u_{n+1}(x) = \nabla u_n(x) + \sum_{Q \in \mathcal{Q}_{n+1}} \frac{du_n}{dn}(a_Q) \nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q)).$$

Lemma 3.

$$\sum_{Q \in \mathcal{Q}_{n+1} : |x - a_Q| > \rho} |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q))| \leq C N^{-d} \delta_{n+1} \rho^{-1}$$

For all $\rho > \text{const} \cdot \delta_{n+1}$ and $x \in \mathbb{R}^{d-1}$ (C depends on a lower bound for $\frac{\rho}{\delta_{n+1}}$)

Proof. The left hand side is $\leq \sum_{|x - a_Q| > \rho} (N\delta_{n+1}^{-1}|x - a_Q|)^{-d}$ by the last part of lemma 1. Lemma 3 follows by considering $\sum \delta_{n+1}^{d-1}|x - a_Q|^{-d}$ as Riemann sum for $\int_{|x| > \rho} |x|^{-d}$, which is justified as long as $\frac{\rho}{\delta_{n+1}}$ does not approach 0. $\square$

Corollary. If δ_{n+1} is small enough, then on $\mathbb{R}^{d-1}$,

$$|\nabla u_{n+1} - \nabla u_n| \leq C K_{n+1} e^{-\beta n} \varepsilon_{n+1}^{-d}$$

Proof. Assume for notational purposes that $x \in Q \in \mathcal{Q}_{n+1}$ - the other case is a little simpler. If δ_{n+1} is small, then by (b) and smoothness on u_n , we know

$$|\nabla u_n(a_{Q'})| \leq 2K_{n+1} e^{-\beta n}$$

for all $Q' \in \mathcal{Q}_{n+1}$. So

$$\begin{aligned} |\nabla u_{n+1}(x) - \nabla u_n(x)| &\leq \sum_{\substack{Q' \in \mathcal{Q}_{n+1} \\ Q' \neq Q}} |\nabla u_n(a_{Q'})| |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'}))| \\ &\quad + |\nabla u_n(a_Q)| |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q))| \\ &\leq 2K_{n+1} e^{-\beta n} \left[\sum_{Q' \neq Q} |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'}))| \right. \\ &\quad \left. + |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q))| \right] \\ &\leq 2K_{n+1} e^{-\beta n} (C + C\varepsilon_{n+1}^{-d}) \end{aligned}$$

where the sum was estimated by lemma 3 and the other term by lemma 1. $\square$

Next we verify the induction hypothesis at stage $n+1$ i.e.

Lemma 4. If A is large then $\left(\int_{V_{n+1}} \left| \frac{du_{n+1}}{dn} \right|^p \right)^{1/p} < A e^{-\beta(n+1)}$

Proof. First we claim the following statement: suppose $\gamma > 0$ is given. Then if δ_{n+1} is small enough we will have for any $x \in Q \in \mathcal{Q}_{n+1}$

$$\sum_{\substack{Q' \in \mathcal{Q}_{n+1} \\ Q' \neq Q}} \left| \frac{du_n}{dn}(a_{Q'}) \right| |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'}))| \leq C N^{-d} \left| \frac{du_n}{dn}(x) \right| + \gamma \quad (*)$$

Proof of ():* Fix $M < \infty$ and split the sum into $\sum_{Q': |x-a_{Q'}| < M\delta_{n+1}}$ and $\sum_{Q': |x-a_{Q'}| > M\delta_{n+1}}$. Then

$$\begin{aligned} \sum_{Q': |x-a_{Q'}| < M\delta_{n+1}} &\leq \sup \left(\left| \frac{du_n}{dn}(a_{Q'}) \right| : |x-a_{Q'}| < M\delta_{n+1} \right) \cdot \sum_{Q' \neq Q} |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x-a_{Q'}))| \\ &< CN^{-d} \sup \left(\left| \frac{du_n}{dn}(a_{Q'}) \right| : |x-a_{Q'}| < M\delta_{n+1} \right) \end{aligned}$$

(by lemma 3 with $\rho = \frac{1}{2}\delta_{n+1}$).

$$< CN^{-d} \left(\left| \frac{du_n}{dn}(x) \right| + \gamma \right)$$

where $\gamma \rightarrow 0$ as $\delta_n \rightarrow 0$ by continuity of $\left| \frac{du_n}{dn} \right|$. Also

$$\begin{aligned} \sum_{Q': |x-a_{Q'}| > M\delta_{n+1}} &\leq \sup \left(\left| \frac{du_n}{dn}(y) \right| : y \in V_{n+1} \right) \cdot \sum_{|x-a_{Q'}| > M\delta_{n+1}} |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x-a_{Q'}))| \\ &\leq 2K_{n+1}e^{-\beta n} \sum_{|x-a_{Q'}| > M\delta_{n+1}} |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x-a_{Q'}))| \end{aligned}$$

(provided δ_{n+1} small enough)

$$\leq CK_{n+1}e^{-\beta n} \cdot M^{-1}$$

by lemma 3 ($\rho = M\delta_{n+1}$). Since M is arbitrary we can make this as small as we want which proves (*).

Now, we will call $Q \in \mathcal{Q}_{n+1}$ *type 1* if $\left| \frac{du_n}{dn}(a_Q) \right| > e^{-4\beta(n+1)}$ and *type 2* otherwise. Fix $Q \in \mathcal{Q}_{n+1}$. Write (for $x \in Q$)

$$\begin{aligned} \frac{du_{n+1}}{dn}(x) &= \frac{du_n}{dn}(a_Q) + \left[\frac{du_n}{dn}(x) - \frac{du_n}{dn}(a_Q) \right. \\ &\quad \left. + \sum_{\substack{Q' \neq Q \\ Q' \in \mathcal{Q}_{n+1}}} \frac{du_n}{dn}(a_{Q'}) \cdot \frac{dF_{\varepsilon_{n+1}}}{dn}(N\delta_{n+1}^{-1}(x-a_{Q'})) \right] + \frac{du_n}{dn}(a_Q) \frac{dF_{\varepsilon_{n+1}}}{dn}(N\delta_{n+1}^{-1}(x-a_Q)) \end{aligned}$$

By (*) and smoothness of $\frac{du_n}{dn}$ the term in brackets may be made less than $CN^{-d} \left| \frac{du_n}{dn}(a_Q) \right| + \gamma$ with γ arbitrarily small. For Q type 1, the bracket term is therefore $< CN^{-d} \left| \frac{du_n}{dn}(a_Q) \right|$ provided γ is chosen right. So lemma 2 applies (we're using here that $CN^{-d} < cN^{-(d-1)}$ for large N) and we get

$$\begin{aligned} \left(\int_Q \left| \frac{du_{n+1}}{dn} \right|^p \right)^{1/p} &< e^{-2\beta} \left| \frac{du_n}{dn}(a_Q) \right| |Q|^{1/p} \\ &< e^{-\frac{3}{2}\beta} \left(\int_Q \left| \frac{du_n}{dn} \right|^p \right)^{1/p} \quad \text{if } \delta_{n+1} \text{ is small} \end{aligned}$$

If Q is type 2 the bracket term is $< CN^{-d} \left| \frac{du_n}{dn}(a_Q) \right| + \gamma$ which may be taken $< e^{-4\beta(n+1)}$. So,

$$\begin{aligned} \int_Q \left| \frac{du_{n+1}}{dn} \right|^p &\leq \left| \frac{du_{n+1}}{dn}(a_Q) \right|^p \int_Q \left| 1 + \frac{dF_{\varepsilon_{n+1}}}{dn}(N\delta_{n+1}^{-1}(x-a_Q)) \right|^p + \int_Q []^p \\ &\leq 2 \cdot e^{-4\beta(n+1)p} |Q| \end{aligned}$$

since (say) the first integral is $< |Q|$ by lemma 2. We conclude

$$\begin{aligned}
\int_{V_{n+1}} \left| \frac{du_{n+1}}{dn} \right|^p &= \int_{\bigcup \{Q \text{ type 1}\}} + \int_{\bigcup \{Q \text{ type 2}\}} \\
&\leq e^{-\frac{3}{2}p\beta} \int_{\bigcup \{Q \text{ type 1}\}} \left| \frac{du_n}{dn} \right|^p + C e^{-4p\beta(n+1)} \sum_{Q \text{ type 2}} |Q| \\
&\leq A^p e^{-p\beta(n+1)} \cdot e^{\frac{1}{2}p\beta} + C e^{-4p\beta(n+1)} \\
&= A^p e^{-p\beta(n+1)} \left(e^{\frac{1}{2}p\beta} + C A^{-p} e^{-3p\beta(n+1)} \right)
\end{aligned}$$

The parenthesis term is < 1 if A was large enough. $\square$

Now we give the conditions on the K_n and ε_n : they should satisfy

- (1) $\sum \varepsilon_{n+1}^{-d} K_{n+1} e^{-\beta n} < \infty$
- (2) $\sum K_{n+1}^{-p} + \varepsilon_{n+1}^{d-1}$ is sufficiently small.

These are clearly compatible e.g. we can take $\varepsilon_n = C^{-1} n^{-2}$, $K_n = C n^{2/p}$ for a suitable large constant C .

Condition (1) implies by the corollary to lemma 2 that the $\hat{u}_n$ converge in C^1 norm on the closed upper half space. Call the limit function f . We have to show $f = \frac{df}{dn} = 0$ on a set of positive measure. But

$$|\{x \in Q(1) : f(x) \neq 0\}| \leq \sum_n |\{x \in Q(1) : u_{n+1}(x) \neq u_n(x)\}|$$

and $\{x : u_{n+1}(x) \neq u_n(x)\}$ is a union of at most $\delta_{n+1}^{-(d-1)}$ discs each of radius $\delta_{n+1} \varepsilon_{n+1}^{1/2}$ (because of $\text{supp } F_\varepsilon$ being contained in $D(0, \varepsilon^{1/2})$). So,

$$\begin{aligned}
|\{x \in Q(1) : f(x) \neq 0\}| &\leq C \sum_n \delta_{n+1}^{-(d-1)} \left(\delta_{n+1} \varepsilon_{n+1}^{1/2} \right)^{d-1} \\
&\leq C \sum_n \varepsilon_{n+1}^{\frac{d-1}{2}}
\end{aligned}$$

Also, $\frac{d\hat{f}}{dn} = 0$ on $\bigcap_n V_n$. Thus

$$\left| \left\{ x \in Q(1) : \frac{d\hat{f}}{dn}(x) \neq 0 \right\} \right| \leq \sum_n |V_n \setminus V_{n+1}|$$

What is the measure of $V_n \setminus V_{n+1}$? If $Q \notin \mathcal{Q}_{n+1}$ but $Q \subset Q' \in \mathcal{Q}_n$, then

$$\frac{1}{|Q|} \int_Q \left| \frac{du_n}{dn} \right|^p > K_{n+1}^p e^{-p\beta n} \quad \text{i.e.} \quad |Q| < K_{n+1}^{-p} e^{-\beta p n} \int_Q \left| \frac{du_n}{dn} \right|^p.$$

So,

$$|V_n \setminus V_{n+1}| \leq K_{n+1}^{-p} e^{\beta p n} \int_{V_n} \left| \frac{du_n}{dn} \right|^p \leq A^p K_{n+1}^{-p}$$

and $\left| \left\{ x \in Q(1) : \frac{d\hat{f}}{dn}(x) \neq 0 \right\} \right| \leq A^p \sum K_{n+1}^{-p}$. That means

$$\left| \left\{ x \in Q(1) : \frac{d\hat{f}}{dn}(x) \neq 0 \right\} \right| + |\{x \in Q(1) : f(x) \neq 0\}| \leq C \sum \varepsilon_{n+1}^{\frac{d-1}{2}} + A^p \sum K_{n+1}^{-p} < 1, \text{ if (2) holds.}$$

So $Q(1) \cap \left\{ x : \frac{d\hat{f}}{dn}(x) = 0 \right\} \cap \{x : f(x) = 0\}$ must have positive measure.

Remarks. 1) To minimize technicalities we did not try to obtain a Hölder estimate on the gradient although this should be possible along the work of [2].

2) The main question remains open, that is, whether it is possible to have harmonic functions C^2 or C^∞ smooth up to the boundary whose gradients vanish on sets of positive measure. A variant of previous construction permits to get a $C^{2+\varepsilon}$ harmonic function $f : \mathbb{R}_+^d \rightarrow \mathbb{R}$ such that f and D^2f vanish on a common boundary of positive measure.

REFERENCES

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